

Nabarun Kar

945-544-1125 | nabarunkar01@gmail.com | [LinkedIn](#) | [GitHub](#) | [Website](#)

EDUCATION

Texas A&M University

Master of Science in Computer Science; GPA: 3.79

College Station, TX

Aug. 2024 – May 2026

St. Thomas' College of Engineering and Technology

Bachelor of Technology in Computer Science and Engineering; GPA: 3.80

Kolkata, West Bengal, India

Aug. 2019 – May 2023

EXPERIENCE

Artificial Intelligence Engineering Intern

June 2025 – August 2025

Texas Capital Bank

Dallas, TX

- Built an end-to-end document-based RAG application using AWS Bedrock, reducing manual document lookup time by over 70%.
- Designed and deployed secure REST APIs for uploading documents up to 50MB to AWS S3 with under 2 seconds average processing time.
- Developed a production-ready backend using Python and FastAPI, supporting concurrent access from 50+ users.
- Automated PDF-to-CSV invoice extraction using Camelot, Tabula, PDFPlumber, and AWS Glue, improving core banking workflows.
- Implemented guardrails for LLM applications to enforce enterprise standards and ensure regulatory compliance.
- Built Grafana dashboards to visualize latency and real-time performance metrics.

Data Engineer

September 2023 – July 2024

Accenture

Kolkata, WB

- Used Kubernetes and Docker for container orchestration and lifecycle management.
- Performed load balancing, troubleshooting, and routine server upkeep.
- Implemented automated cluster startup and configuration changes using Python, improving efficiency by 50%.

Java Intern

March 2023 – May 2023

LTIMindtree

Kolkata, WB

- Built a vendor warehousing application using Spring Boot and microservices.

PROJECTS

Filmoid | *Python, FastAPI, React, PostgreSQL, Docker, Surprise (scikit-surprise)*

July 2026 – Present

- Developed a full-stack movie recommendation application using React and a FastAPI REST API backend.
- Built a collaborative-filtering recommendation engine using a trained Surprise SVD model.
- Implemented JWT authentication, persistent user ratings, and recommendation history using PostgreSQL.
- Containerized and deployed the ML-backed backend with Docker on Hugging Face Spaces.

CrushedPlay | *React, TypeScript, Node.js, Express, WebRTC, WebSockets*

July 2026 – Present

- Built a peer-to-peer synchronized watch-party application that streams multi-gigabyte video files directly between browsers using WebRTC DataChannels.
- Implemented WebSocket-based signaling and real-time playback synchronization with a decoupled React frontend and Node.js/Express backend.
- Streamed 64KB video chunks directly into the browser's Origin Private File System (OPFS), bypassing RAM limitations for large files.

TECHNICAL SKILLS

Languages: Python, Java, C++, JavaScript, TypeScript, SQL

Frameworks & Libraries: React, FastAPI, Express.js, Spring Boot, TensorFlow, PyTorch, Scikit-learn, SQLAlchemy, Pandas, NumPy, WebRTC, WebSockets

Cloud & DevOps: AWS (S3, Glue, Bedrock), Docker, Kubernetes, Jenkins, Git, CI/CD, Vercel, Hugging Face

Databases & Backend: PostgreSQL, MongoDB, DynamoDB, Node.js, REST APIs

ACHIEVEMENTS

Winner, E.P.I.C IT Challenge S4 (TVSCredit): *National hackathon winner among 17,901 participants*

Unstop Top 70 E-School Leaders (2023): *Recognized as one of the top engineering talents nationwide*